gentle: They grow so quickly, so suddenly, that, if you encounter one, your net distortion is finite, not infinite, at the moment you hit the singularity.

In *Interstellar*, Romilly tells Cooper about these gentle singularities: "I have a suggestion for your return journey [from Mann's planet]. Have one last crack at the black hole. Gargantua's an older, spinning black hole. [It has] what we call a gentle singularity." "Gentle?" Cooper asks. "They're hardly gentle, but their tidal gravity is quick enough that something crossing the horizon fast might survive." Cooper, lured by this conversation and the quest for quantum data, later plunges into Gargantua (Chapter 28). It's a brave plunge. He can't know in advance whether he'll survive. Only the laws of quantum gravity know for sure. Or the bulk beings . . .

We've now laid the extreme-physics foundations for *Interstellar*'s climactic scenes, so let's turn to the climax.

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45 More precisely, the locations of the mirrors' centers of mass.

46 In 1955, John Wheeler pointed out the likely existence of quantum foam, with wormhole sizes 10^{-35} meters: 10 trillion trillion times smaller than an atom; the so-called *Planck length*.

47 The thing he simulated was actually something called a scalar wave, but that is an irrelevant technicality. A few years later Andrew Abrahams and Chuck Evans at the University of North Carolina repeated Choptuik's simulations using a gravitational wave and got the same result: a naked singularity.

48 Israel and Poisson gave this singularity the name *mass inflation singularity*, and that is the name that physicists have used ever since. I prefer *infalling singularity* and use that name in this book.